

CLASS ASSIGNMENT => 01

1) Write Five Advantages of JAVA and Compare Java with C/C++.

=> Five Advantages of JAVA:

1. Platform Independent:

=> Java follows the principle "write Once, Run Anywhere (WORA)."

=> Java programs run on any operating system that has the Java Virtual Machine (JVM).

2. Object-Oriented:

=> Java is based on object-oriented programming concepts like class, objects, inheritance, polymorphism, encapsulation, and abstraction.

3. Secure:

=> Java provides built-in security features such as bytecode verification, class loaders, and no direct pointer manipulation.

4. Robust:

=> Java has strong memory management, exception handling, and automatic garbage collection, making programs more reliable.

5. Multithreading Support:

=> Java supports multiple threads, allowing several tasks to run simultaneously and improving application performance.

2. Comparison of JAVA with C/C++

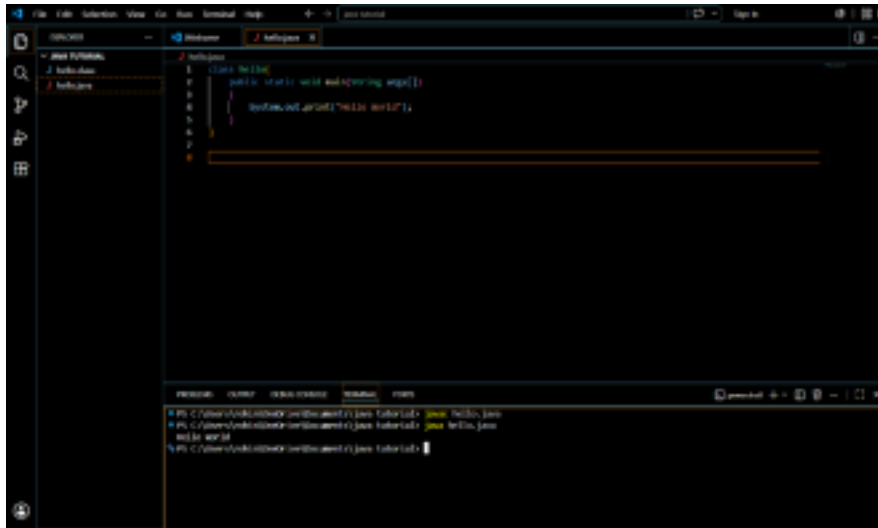
Feature	JAVA	C/C++
Platform	Platform Independent	Platform dependent
programming style	pure Object-oriented (mostly)	procedural + Object-oriented (C++)
Memory management	Automatic Garbage Collection	Manual memory management.
Pointers	No Direct pointers	Supports pointers
Security	More Secure	Less Secure
Compilation	Compiles to Bytecode (JVM)	Compiles to machine Code
Multiple Inheritance	Through Interfaces	Supports in C++
Portability	Highly portable	Less portable.

ASSIGNMENT 1 REG.NO: 192424413 JAVA INSTALLATION COMPLETED

```
C:\Users\rohin>javac --version
javac 17.0.12

C:\Users\rohin>java --version
java 17.0.12 2024-07-16 LTS
Java(TM) SE Runtime Environment (build 17.0.12+8-LTS-286)
Java HotSpot(TM) 64-Bit Server VM (build 17.0.12+8-LTS-286, mixed mode, sharing)
```

HELLO WORLD PROGRAM:



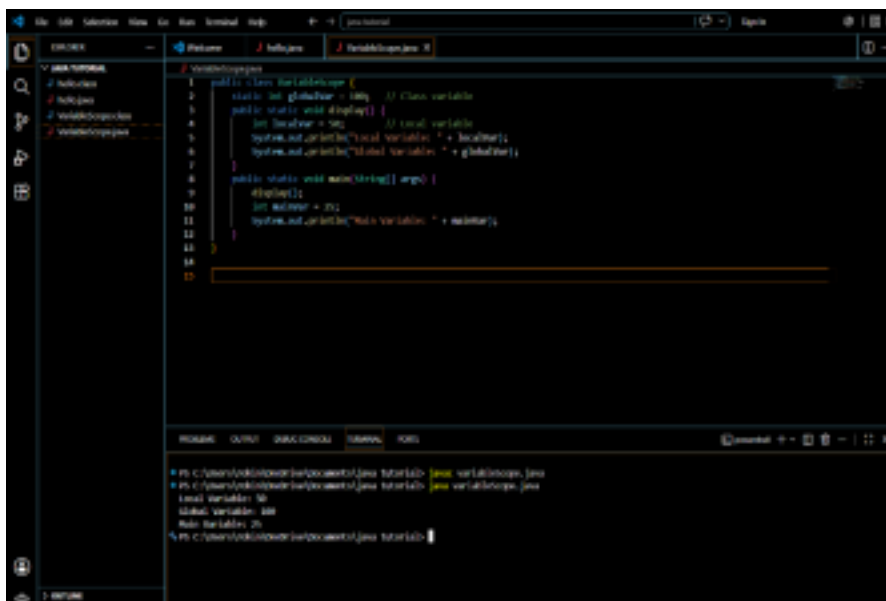
The screenshot shows an IDE with a project named 'HelloWorld'. The main file is 'Hello.java', which contains the following code:

```
1 class Hello {
2     public static void main(String args[]) {
3         System.out.println("Hello world!");
4     }
5 }
6
```

The output window at the bottom shows the execution results:

```
PS C:\Users\rohin\source\repos\HelloWorld> java HelloWorld
Hello world
```

VARIABLESCOPE:



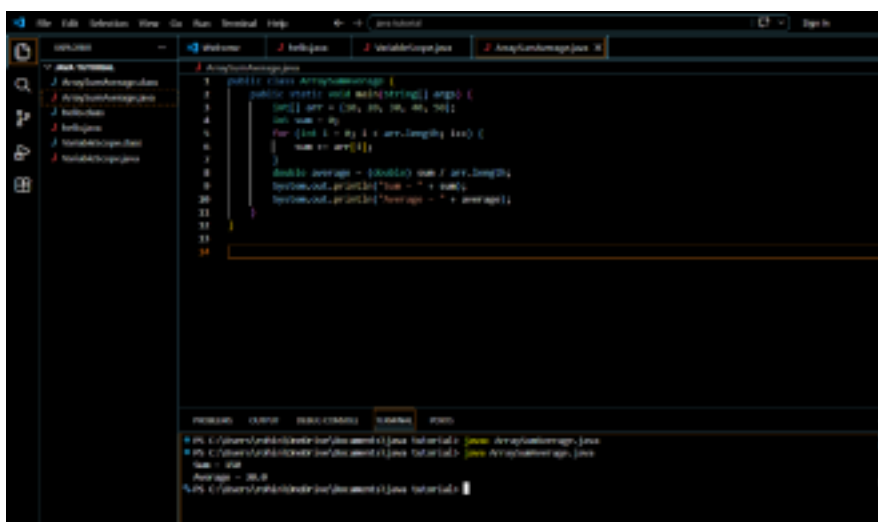
The screenshot shows an IDE with a project named 'VariableScope'. The main file is 'VariableScope.java', which contains the following code:

```
1 public class VariableScope {
2     static int globalVar = 10; // Class variable
3     public static void display() {
4         int localVar = 20; // Local variable
5         System.out.println("Local variable: " + localVar);
6         System.out.println("Global variable: " + globalVar);
7     }
8     public static void main(String[] args) {
9         display();
10        int localVar = 30;
11        System.out.println("Local variable: " + localVar);
12    }
13 }
14
```

The output window at the bottom shows the execution results:

```
PS C:\Users\rohin\source\repos\VariableScope> java VariableScope
Local variable: 30
Global variable: 10
Local variable: 20
Global variable: 10
```

ARRAYSUMAVERAGE:



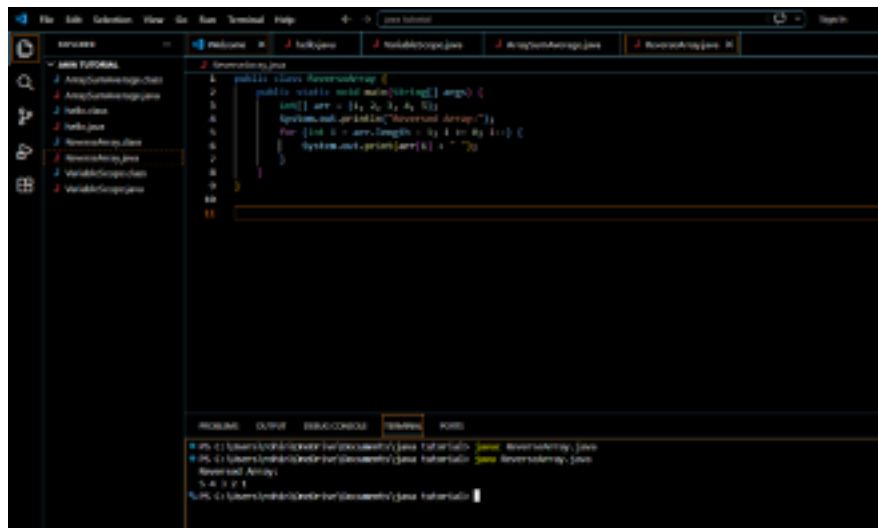
The screenshot shows an IDE with a project named 'ArraySumAverage'. The main file is 'ArraySumAverage.java', which contains the following code:

```
1 public class ArraySumAverage {
2     public static void main(String[] args) {
3         int[] arr = {10, 20, 30, 40, 50};
4         int sum = 0;
5         for (int i = 0; i < arr.length; i++) {
6             sum += arr[i];
7         }
8         double average = (double) sum / arr.length;
9         System.out.println("Sum = " + sum);
10        System.out.println("Average = " + average);
11    }
12 }
13
```

The output window at the bottom shows the execution results:

```
PS C:\Users\rohin\source\repos\ArraySumAverage> java ArraySumAverage
Sum = 150
Average = 30.0
```

REVERSEARRAY:

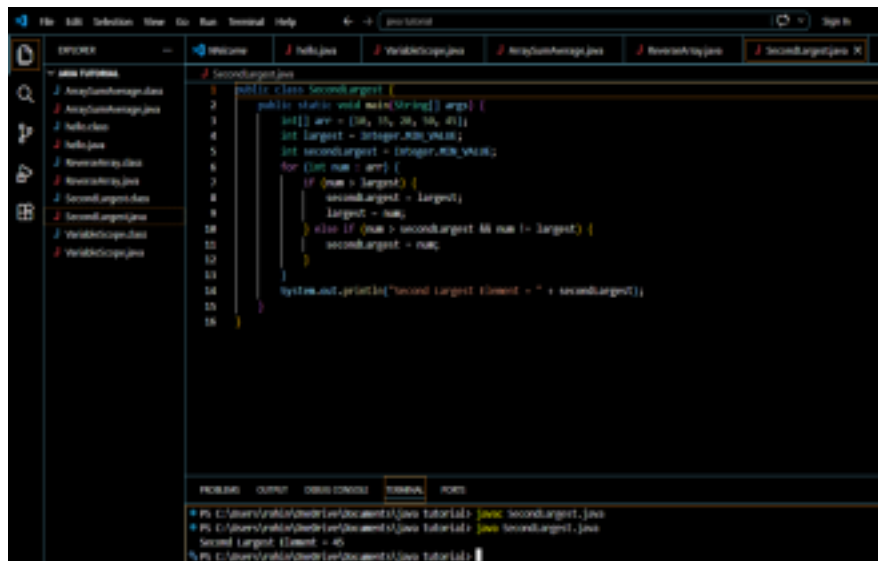


```
1 public class ReverseArray {
2     public static void main(String[] args) {
3         int[] arr = {1, 2, 3, 4, 5};
4         System.out.println("Reverse array:");
5         for (int i = arr.length - 1; i >= 0; i--) {
6             System.out.print(arr[i] + " ");
7         }
8     }
9 }
10
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
* C:\Users\rishu\OneDrive\Documents\java tutorial> javac ReverseArray.java
* C:\Users\rishu\OneDrive\Documents\java tutorial> java ReverseArray.java
Reverse Array:
5 4 3 2 1
* C:\Users\rishu\OneDrive\Documents\java tutorial>
```

SECONDLARGEST:



```
1 public class SecondLargest {
2     public static void main(String[] args) {
3         int[] arr = {10, 15, 20, 30, 40};
4         int largest = Integer.MIN_VALUE;
5         int secondlargest = Integer.MIN_VALUE;
6         for (int num : arr) {
7             if (num > largest) {
8                 secondlargest = largest;
9                 largest = num;
10            } else if (num > secondlargest && num != largest) {
11                secondlargest = num;
12            }
13        }
14        System.out.println("Second largest element = " + secondlargest);
15    }
16 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
* C:\Users\rishu\OneDrive\Documents\java tutorial> javac SecondLargest.java
* C:\Users\rishu\OneDrive\Documents\java tutorial> java SecondLargest.java
Second largest element = 30
* C:\Users\rishu\OneDrive\Documents\java tutorial>
```